Distribution	Parameters	Equation	Mean	Variance	Code
Bernoulli	p	$P[X=1]=p$ $P[X=0]=1-p$	p	$p(1-p)$	bernoulli.pmf()
Binomial	n, p	$P[X=k]={}^{n}C_{k}p^{k}(1-p)^{n-k}$	np	$np(1-p)$	binom.pmf()
Geometric	p	$P[X=k]=(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	geom.pmf()
Poisson	λ	$P[X=k]=\frac{\lambda^{k}e^{-\lambda}}{k!}$	λ	λ	poisson.pmf()
Exponential	λ	$F(x)=1-e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	expon.cdf()
Normal	μ, σ	$f(x)=\frac{1}{2\sigma^2}e^{\frac{(x-\mu)^2}{\sigma^2}}$	μ	σ	norm.cdf()
Log normal	μ, σ	$\log X \sim$ Gaussian	$exp(\mu+\sigma^2/2)$	$e^{(\sigma^2)-1}e^{2\mu+\sigma^2}$	lognorm.cdf()